

## Factoring Large Numbers

🔖 Tags:

Time Limit: 1 s

Memory Limit: 128 MB

Submission: 112

AC: 62

Score: 98.72

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Codes

AI Code Tips

### Description

One of the central ideas behind much cryptography is that factoring large numbers is computationally intensive. In this context one might use a 100 digit number that was a product of two 50 digit prime numbers. Even with the fastest projected computers this factorization will take hundreds of years. You don't have those computers available, but if you are clever you can still factor fairly large numbers.

### Input

The input will be a sequence of integer values, one per line, terminated by a negative number. The numbers (other than the terminator) will be greater than 0 and less than  $2^{62}$ .

### Output

Each positive number from the input must be factored and all factors (other than 1) printed out. The factors must be printed in ascending order with 4 leading spaces preceding a left justified number, and followed by a single blank line.

### Samples

| input                                                                | Copy |
|----------------------------------------------------------------------|------|
| 90<br>1234567891<br>18991325453139<br>12745267386521023<br>-1        |      |
| output                                                               | Copy |
| 2<br>3<br>3<br>5<br><br>1234567891<br><br>3<br>3<br>13<br>179<br>271 |      |

1381  
2423  
  
30971  
411522630413

## Source

19 June, 2011 - Waterloo local contest

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HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 14:59:02